

WELL COMPLETION REPORT (SYNTHETIC)

OIL-BGJ-006 - Baghjan-006

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000069
Field / District	Baghjan / Tinsukia
Block	OALP-IV-03
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2003-11-13 / 2004-01-07
Measured / TVD	3145 m / 3101 m
Trajectory	Vertical; inc 5.1 deg; az 318.7 deg
Formation Interval	1781-2941 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	3598 psi
Fracture Gradient	14.94 ppg
Mud System	Glycol-PHPA WBM
NWIS Well Risk	High (58.6/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	786	Surface
9 5/8 in casing	1824	Intermediate
7 in liner/casing	2767	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2003-11-23 729 m	Mud Loss	High	Intermittent circulation losses developed while drilling ahead. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Use conservative hydraulics when re-entering equivalent loss-prone interval.
2003-12-04 1333 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Use offset torque trend to set adaptive operating envelope.
2003-12-04 1358 m	Cementing Issue	High	Cement returns were below planned percentage. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.
2003-12-10 1662 m	Pressure Spike	High	Pressure trend exceeded recent moving baseline. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2003-12-14 1862 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Apply approved freeing procedure and avoid excessive overpull. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2003-12-14 1912 m	Kick	High	Drilling break followed by positive flow check. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Maintain mud-weight window using offset pressure evidence.
2003-12-27 2547 m	Stuck Pipe	High	Rotary movement deteriorated progressively. Mitigation: Reduce ROP and increase circulation before high-angle interval. Lesson: Use offset stuck-pipe depth to increase circulation and reaming frequency.

Completion Summary

The synthetic well reached 3145 m MD in the Barail interval. The NWIS knowledge record links 10 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.